

CStats - Lab #1

February 24, 2026

Exercise 1.

Suppose that a school has 20 classes: 16 with 25 students in each, three with 100 students in each, and one with 300 students, for a total of 1000 students.

- (a) What is the average class size?
- (b) Select a student randomly out of the 1000 students. Let the random variable X equal the size of the class to which this student belongs, and define the probability mass function of X .
- (c) Find $\mathbb{E}(X)$, the expected value of X . Does this answer surprise you ?

Exercise 2.

A coin comes up heads with probability p and tails with probability $1 - p$. We do not know p . We flip the coin 100 times and observe X heads. Derive $\hat{p}_{ML}(X)$, the ML estimator for p .

Exercise 3.

Let $X_1, X_2, \dots, X_n$ be i.i.d. samples of r.v. X with unknown finite mean and variance. Let $\bar{X}$ denote the sample mean. Define

$$\bar{S}^2 \equiv \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Prove that $\bar{S}^2$ is **not** an unbiased estimator of $\text{Var}(X)$. Follow these steps:

- (a) Prove that

$$\mathbb{E}[\bar{S}^2] = \frac{1}{n} \sum_{i=1}^n \text{Var}(X_i - \bar{X}).$$

- (b) Show that

$$\text{Var}(X_i - \bar{X}) = \frac{n-1}{n} \text{Var}(X).$$

- (c) Combine (a) and (b) to show that

$$\mathbb{E}[\bar{S}^2] = \frac{n-1}{n} \text{Var}(X).$$

Exercise 4.

Team A has probability p of beating team B. We do not know p , but we can see that in the last 10 games played between A and B, team A won seven games and team B won three games. Assume that every game has a unique winner and that games are independent. Based on this information, formulate and compute the ML estimator for p .

Exercise 5.

A casino suspects that one of its roulette tables may be malfunctioning. To investigate, the casino records the net gain (in CHF) per round for a random sample of $n = 64$ rounds played by different customers. The sample yields:

$$\bar{X} = -2.5 \text{ CHF}, \quad s = 4.0 \text{ CHF}.$$

- (a) Using the Central Limit Theorem, explain why we can assume that $\bar{X}$ follows an approximately normal distribution, regardless of the underlying distribution of individual round outcomes.
- (b) Construct a 95% confidence interval for the true mean net gain per round μ . (Use $z_{0.025} = 1.96$.)
- (c) Under normal operating conditions, the expected net gain for the player is -4 CHF per round (i.e., the house edge). The casino wants to test whether the table is deviating from this expected value. Formulate a two-sided hypothesis test at significance level $\alpha = 0.05$ and state your conclusion.
- (d) If the casino wants the 95% confidence interval to have a total width of at most 1 CHF (i.e., margin of error ≤ 0.5 CHF), what is the minimum number of rounds it must observe? Assume the same standard deviation $s = 4.0$.